
ME 452: Robotics and Control

Course Code	ME 452
Course Name	Robotics and Control
L-T-P-C	2-1-0-3
Prerequisites	ME 309 Theory of Machine or consent of faculty
Intended for	B-Tech final year/pre-final, MS.
Distribution	Elective
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Contents

- **Introduction to Robotics:** Basic definitions, mechanism, degree of freedom, classification and specifications of Robots, Industrial Robots, sensors, controller, actuator. (4 Lectures)
- **Robot Kinematics:** Position and orientation of links, Coordinate transformation, d-h parameters, joint variable and position of end effectors, inverse kinematic analysis. Velocity analysis - Jacobian. Static force analysis. (10 Lectures)
- **Trajectory Planning:** joint space trajectory planning, cubic polynomial path generation, obstacle avoidance (5 Lectures)
- **Robot Dynamics:** Derivation of dynamics equation based on Newton Eulers formulation and Lagrangian formulation. (9 Lectures)
- **Robot Control:** Actuators- hydraulic, pneumatic, electric motors, Sensors- position, velocity, proximity, force and pressure, Position control- Proportional-Integral-Derivative control, servo compensation. Force control - Impedance control, hybrid control (force + position control), introduction to nonlinear control of manipulators. (14 Lectures)

Text Books

1. Craig John J., **Introduction to robotics: Mechanics & Control**, 3rd Edition, Pearson. 2008.
2. Tsuneo Yoshikawa, **Foundations of Robotics - Analysis and Control**, 1990.

References

1. Niku Saeed B., **Introduction to Robotics: Analysis, Systems, Applications**, 2nd edition, Wiley, 2011.
2. Richard K. Lafter, T. Chmiewski, M. Nigin, **Robotics Engineering**, PHI India
3. Michel P Grover, et. al., **Industrial Robotics**, McGraw Hill
4. Subir Kumar Saha, **Introduction to robotics**, Mc Graw Hill